Background Removal for Multiplex Imaging in Tissue Samples

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Abstract. Multiplexed imaging techniques, such as immunohistochemistry (IHC), imaging mass cytometry (IMC), and cytoclic immunofluorescence (CyCIF), allow researchers to simultaneously measure multiple biological markers in tissue samples. However, analyzing these complex, high-dimensional datasets can be challenging. To make sense of the data, automated tools are needed to perform tasks such as background removal, noise reduction, cell segmentation, feature identification, and spatial analysis. The focus of my project is to explore computer vision-based methods for interpreting multiplexed imaging data. Given the constraints of this course's term project, the scope has been narrowed to the critical first step in the pipeline, the removal of background images from the slides. This step, which involves eliminating non-tissue regions or nonspecific signals, is a common pre-processing task in histopathology image analysis. There are several approaches, both traditional and deep learning based, designed to address background removal in pathology images. For my project, I will implement and compare three open source background removal source code from GitHub repositories, and compare the results to find an optimum solution for the removing noises of multiplex images.

1 Introduction

Multiplexed imaging technologies, such as immunohistochemistry (IHC), imaging mass cytometry (IMC), and cytoclic immunofluorescence (CyCIF), have completely changed how we study tissues at the single-cell level. These advanced methods allow researchers to measure many biomarkers simultaneously in tissue sections, providing deep insights into tissue structure, cellular characteristics, and how different cell types interact. For example, in cancer research, multiplexed imaging helps map the tumor microenvironment (TME) by identifying different cell types. This is crucial to understand how tumors grow, spread, and respond to treatments [5]. Similarly, in immunology, multiplexed imaging enables the study of how immune cells interact within tissues, in autoimmune disorders, infections, and cancer immunotherapy [7]. While the potential of multiplexed imaging is enormous, analyzing the resulting data is a significant challenges. A critical challenges in these image analysis is background noise, which may arises due to

- Autofluorescence of Tissue Components. Blood cells and extracellular matrix components (e.g. collagen, elastin) naturally emit fluorescence, causing nonspecific signal interference.
- or by Residual Background from Quenching and Restaining Cycles. In sequential multiplex staining, quenching reagents do not fully remove prior fluorescence signals, leading to cumulative background noise. This background contamination reduces signal specificity, affects quantification accuracy, and complicates downstream analyses, ultimately limiting the reliability of results in cancer research and biomarker discovery.
- Existing Software Limitations: Current manual analysis methods are slow, error-prone, and not scalable for large datasets [3]. There is a critical need for automated, scalable, and reproducible analysis pipelines that can process this data efficiently with minimal manual input.

  The common image analysis tools, such as QuPath and ImageJ, work well for white background images but struggle with complex tissue backgrounds where different cell types and staining intensities overlap.

  To address these challenges, several computational tools have been developed. For example, PHLEX, a deep learning-based segmentation method [5], and IBEX, an iterative labeling and bleaching method [7], both improve the efficiency of analyzing multiplexed imaging data. Additionally, mplexable, a Python library, enhances the CyCIF technique by optimizing signal removal and antibody specificity [3]. Moreover, SIMPLI offers a flexible, technology-agnostic software solution for spatially resolved tissue phenotyping and single-cell analysis [2]. However, despite these advancements, the tools available are often complex, require specialized expertise, and may not be compatible with all imaging methods. This project will focus on deploying and evaluating the cutting-edge algorithms like deep-imcyto, IBEX, mplexable, and SIMPLI. The main goal is to assess the strengths and weaknesses of each method in the context of multiplexed imaging, particularly in their ability to perform cell segmentation, phenotyping, and spatial analysis.

There are both traditional and deep learning methods focused on background removal in IHC/pathology images, with references to research papers and links to open-source implementations (including those for multiplex imaging):

- PyHIST (Whole-Slide Tissue Segmentation) [6] An open-source tool that uses color processing and thresholding to generate tissue masks on whole slide images. PyHIST separates tissue from blank background and then tiles the tissue regions for analysis. The code is available on GitHub for easy integration into pipelines.
- PENGUIN (Multiplex Image Preprocessing) [9] A method designed for multiplex spatial proteomics images (e.g. Imaging Mass Cytometry). PENGUIN uses scaling, adaptive thresholding, and percentile filtering to subtract background noise and hot pixels in multi-channel immunofluorescence data.It's

a fast, non-ML approach that preserves true signal intensity. The pipeline is openly released on GitHub for reproducibility.

- CNN Tissue/Background Segmentation (Resolution-Agnostic) [1] This study developed fully convolutional networks to distinguish tissue from background in whole-slide images. The proposed resolution-agnostic model outperformed traditional thresholding methods, achieving about 97–98% Dice similarity in segmenting tissue vs background.The authors provide their trained model and inference code openly on GitHub.
- U-Net Architectures for Background Removal. [8] A comparative study that evaluated multiple U-Net variants (with backbones like EfficientNet, ResNet, etc.) for histopathology background removal. They treated tissue-vs-background segmentation as a binary mask prediction on WSIs, and found EfficientNet-B3 and MobileNet U-Nets performed best (99% sensitivity and specificity) on The Cancer Genome Atlas data. (The experiment's source code and labeled dataset were used to benchmark U-Net models; results inform which architectures work best for this task.)
- DeepLIIF (Deep Learning for IHC to Multiplex IF) [4] A multitask deep learning framework that performs stain deconvolution and background removal on IHC images, effectively translating brightfield IHC into clean multichannel immunofluorescence. DeepLIIF separates the DAB stain from hematoxylin (producing "clean" fluorescence-like channels) and simultaneously segments cells. It was trained on co-registered IHC and multiplex IF images and can isolate true signal from IHC background artifacts. The authors have released the code (with pretrained models and even a cloud platform) for others to use and reproduce results.

Each of these papers provides insights into background removal for pathology images. Traditional methods like thresholding (as in PyHIST and PENGUIN) are fast and interpretable, while deep learning methods (U-Net variants, DeepLIIF, etc.) often yield higher accuracy and can handle complex multiplex data. All the mentioned approaches come with open-source implementations (on GitHub or similar), facilitating their adoption in multiplex IHC/IF image analysis workflows.

We will use the code from three papers to automatically detect and remove autofluorescence and quenching artifacts from mIHC images, while preserving the integrity of the true staining signals. The goal is to enhance the contrast of these true signals, making biomarker detection in tissue sections clearer and more accurate. Additionally, we will ensure that the approach works across various tissue types and staining protocols, making it adaptable to a wide range of datasets.

The results from this project will be used in a lab at Mount Sinai Hospital, where a team is conducting breast cancer research. Ultimately, the project aims to contribute to the broader effort of automating the analysis of complex tissue data for biomedical research.

2 Research Plan

This project will focus on deploying and comparing three GitHub implementations of state-of-the-art algorithms for background removal in multiplexed images. The project is expected to fit a 7-week timeline and will involve the following key tasks: algorithm deployment, performance evaluation, and comparative analysis.

2.1 Tasks and Methodology

- **Week 1** – Literature Review and Selection of Papers During the first week, I will thoroughly explore the literature to review different algorithms developed for background removal in multiplexed imaging. My focus will be on understanding the core principles behind each method, which will help me assess their relevance and effectiveness. I will carefully select three papers that offer open-source implementations specifically designed to address background removal in multiplexed images. The main goal is to prepare the system for deployment in the upcoming stages.

- **Week 2** – Data Collection and Preprocessing In this week, I will collect publicly available multiplexed imaging datasets, with a particular emphasis on mIHC images that contain background noise, to ensure that the background removal process is meaningful. I will standardize the image formats, normalize the sizes of the images, and verify that the datasets are compatible with the algorithms I have chosen. This step is essential to ensure that everything is properly prepared for the next stages of the project.

- **Week 3-5** – Deploying the Open Source Code Over the next three weeks, I will focus on deploying the open-source code from the three selected papers, spending one week on each. I will begin by setting up the first algorithm and testing how well it performs in removing background noise. Once that is successfully deployed and functioning, I will proceed with the second and third algorithms, making sure to give each method the attention it requires for thorough evaluation.

- **Week 6** – Testing and Evaluation Once the code is deployed, I will test each algorithm thoroughly using a standardized set of multiplexed tissue images. During this phase, I will assess how effectively each method removes background noise, its computational efficiency, and how well it handles larger datasets. I will also evaluate important performance metrics, such as segmentation accuracy and processing time, to gain a comprehensive understanding of how each algorithm performs.

- **Week 7** – Comparative Analysis and Final Report In the final week, I will compare the results of the three algorithms, analyzing the strengths and weaknesses of each method. I will then write a comprehensive report that summarizes my findings, including a comparison of their performance and recommendations for which algorithm is best suited for different types of multiplexed imaging data.

3 Status of work done so far

I have already done the literature review and identified the relevant GitHub repositories for the selected algorithms, but I prefer to study more about it. I have also begun working with the GitHub code from the paper [3] and have accessed it through the github project. While I have made some progress, I need more time to fully implement and test it. The next steps involve deploying the code for mplexable and using it's background removal tool, preparing the system to implement the remaining two algorithms.

4 Conclusion

In this project, I will test various algorithms for background removal in multiplexed imaging, focusing on overcoming the common challenge of background noise in pathology images. Background noise, whether from autofluorescence, quenching artifacts, or residual staining, can significantly impact the accuracy of downstream analyses. By comparing traditional and deep learning-based methods, including tools like PyHIST, PENGUIN, and U-Net variants, this project aims to identify effective solutions for cleaner, more accurate tissue imaging. The ultimate goal is to improve biomarker detection and enable more reliable interpretations of complex tissue data in cancer research and other biomedical applications. With the code from multiple open-source repositories, this project will contribute valuable insights into automating background removal, ensuring reproducibility and scalability across various datasets and tissue types. The results will be used in a lab at Mount Sinai Hospital, assisting a team of researchers working on breast cancer, and will play a part in advancing automated image analysis for scientific and clinical purposes.

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